

Cameron Hastings

North Andover, MA | cjhastings@wpi.edu | 978-884-7104 | linkedin.com/in/cameronjhastings/

Education

Worcester Polytechnic Institute, School of Arts & Sciences , Worcester, MA	Expected May 2028
Bachelor of Computer Science • Dean's List	
St. John's Preparatory School , Danvers, MA	May 2024
National Honor Society • Computer Science Academic Award	

Technical Projects

<i>WPI IQP Project with Banco de Alimentos Panama</i>	March 2025 - Present
- Engineering an end-to-end Python processing pipeline using Pandas, Jinja2, and Playwright to extract raw Excel donor metrics, transforming them into customized PDF traceability reports complete with dynamic geospatial mapping and precise regional weight distribution data. - Developing a targeted Python web scraper to aggregate local supermarket retail pricing into Salesforce-formatted CSV payloads enabling automated fair-market value tracking. - Leading the operational migration from dual inventory platforms to a centralized Salesforce inventory architecture, while simultaneously conducting technical feasibility research on pulp extraction and dehydration methods to significantly extend the shelf life of perishable foods.	
<i>Operating Systems Simulations in C</i>	Spring 2026
- Concurrent Package Processing: Developed a multithreaded package-processing simulator using pthreads, semaphores, and mutexes to coordinate access to shared resources, prevent race conditions, and track per-station throughput under deterministic workloads. - Virtual Memory Manager: Engineered a memory simulator supporting paging, per-process page tables, address translation, and memory protection; designed a deterministic page replacement and disk-swapping mechanism for multi-process execution under memory constraints. - Process Scheduler: Implemented FIFO, Shortest Job First, and Round Robin scheduling policies using C and custom linked lists; evaluated policy performance by computing response, turnaround, and wait times across deterministic workloads.	
<i>Graphing Calculator in Java – Equation Parser</i>	Spring 2025
Developed a program that uses object-oriented design to parse strings of algebraic expressions into structured groupings of expressions and dynamically calculate and render equation graphs in a custom user interface.	
<i>Database Parsing in Java – IMDB Graph Traversal</i>	Spring 2025
Parsed the IMDB database in a navigable binary search tree using object-oriented programming; built and tested various pathing methods between actors and films to identify the algorithm that produced the shortest path.	
<i>Cybersecurity Project – High School Portal IDOR Vulnerability</i>	March 2024
Identified an Insecure Direct Object Reference vulnerability in a student portal and presented findings to IT administrators for mitigation.	

Leadership & Activities

WPI Social Committee, Worcester, MA	November 2024 - Present
Vice President (2026 - Present) Special Events Chair (2025 - 2026) Engagement Chair (2024 - 2025)	
- Execute Vice Presidential duties for the primary campus programming organization, overseeing committee leadership, executive organization, and multi-event planning across campus. - Direct end-to-end production for "Big Show," WPI's flagship spring concert for 1,000+ students, managing artist contracts, artist relations, as well as internal and external vendor operations. - Synchronize 9+ campus and municipal entities including Campus Police, Security, EMS, Facilities, Student Activities, Catering, and Athletics & Rec to execute seamless event security, crowd management, and venue operations.	
Sigma Pi Fraternity, Gamma Iota Chapter, Worcester, MA	March 2025 – Present
House Manager	
- Manage day-to-day operations and safety compliance for an 80+ member facility, leading deep cleaning initiatives and collaborating with the alumni board to ensure successful completion of annual health, safety, and fire inspections. - Designed a preference-based assignment system to allocate brother work duties and parking spots, maintaining strict accountability and operational readiness across the house. - Coordinate routine repairs and multi-thousand-dollar facility upgrades with the alumni board while serving as the primary liaison to WPI administration for campus compliance and life-safety drills.	
WPI Masque Theater, Worcester, MA	August 2025 - May 2026
Head Carpenter	
Produced CAD models to generate precise material estimates and coordinated crew operations across scenic build, load-in, and strike processes under strict deadlines.	

Experience

Summer at St. John's , Danvers, MA	June 2024 – August 2026
<i>Senior Camp Counselor</i>	
Led daily operations and supervised a team of counselors for 100+ campers while managing high-pressure situations and maintaining state protocols.	

Additional Skills & Experience

Programming & Systems: Java, C++, C, Python, MySQL, Object-Oriented Programming, Multithreading & Concurrency (pthreads, semaphores, mutexes), Data Structures, Memory Management Concepts	
Software & Tools: Git/GitHub, Kali Linux, VS Code, IntelliJ, Microsoft Suite, Google Workspace	